Can clustering-based market segmentation improve the prediction of Airbnb listing demand across California?

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Introduction

The dataset used in this project comes from Airbnb listing data collected across California. It's called "Inside AirBnb." The data represents a snapshot of the active Airbnb market as of September 2025. Inside Airbnb periodically captures these snapshots to show what the platform looks like at a specific point in time. Because of this, the dataset reflects listings that were visible and active on the platform during that period rather than a full historical record.

The dataset includes listings from eight California markets, including Los Angeles, Oakland, Pacific Grove, San Diego, San Francisco, San Mateo, Santa Clara County, and Santa Cruz County. These locations represent a mix of large metropolitan tourism markets and smaller coastal regions, which helps provide a broader view of the Airbnb market across California.

In total, the dataset contains over 81,000 listings and 78 columns of information, making it a fairly large dataset for analyzing patterns in the short-term rental market.

However, it is important to note some limitations of the data. Because the dataset is built from publicly available listings, it may not capture listings that are hidden, inactive, or extremely new on the platform. As a result, the dataset should be interpreted as an approximation of the Airbnb market rather than a perfectly complete record. Overall, this dataset provides a large, real-world snapshot of Airbnb activity in California, which makes it useful for analyzing patterns in the short-term rental market.

Airbnb is an online marketplace that allows property owners to rent out short-term accommodations to travelers. Since its launch in 2008, Airbnb has grown into one of the largest platforms in the global hospitality industry and has significantly influenced how people travel and how property owners monetize their homes.

Since Airbnb operates as a digital platform, large amounts of listing information are publicly visible. Researchers and data scientists often compile this information to study trends in short-term rental markets. The dataset used in this project is part of this broader effort to make Airbnb market data accessible for analysis and research.

These datasets are typically collected through publicly available listing information on the Airbnb website and are aggregated by open data projects and data repositories such as Kaggle or Inside Airbnb. The goal of these datasets is to allow researchers, analysts, and policymakers to better understand how short-term rentals operate, how they impact local housing markets, and how demand for short-term accommodations changes across regions.

California is an especially interesting region to study because it contains some of the largest and most competitive Airbnb markets in the United States, including cities such as Los Angeles, San Diego, and San Francisco. These markets attract both leisure and business travelers and include a

wide variety of property types and hosting styles.

Overall, this dataset provides a real-world snapshot of the short-term rental ecosystem in California, making it useful for analyzing trends in listing performance, market demand, and platform dynamics within the sharing economy.

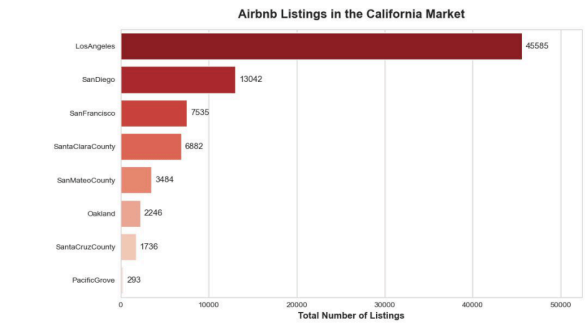
Data Cleaning and Exploratory Data Analysis

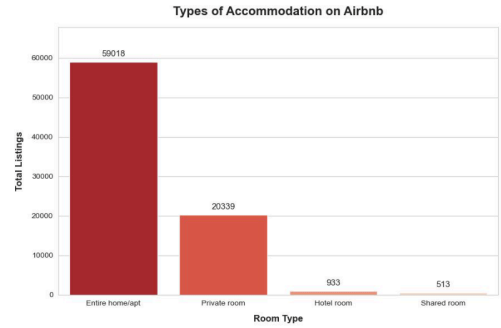
The first step of our project was to clean our datasets in order to ensure that our analysis is accurate and reliable. We began by consolidating eight Airbnb datasets regarding different California markets into one unified dataframe while preserving market identity through a location variable. Next we selected the most relevant features for analysis, such as host characteristics, property details, pricing, availability, and review metrics. To address any missing data, we created a binary indicator to capture whether a listing had reviews and imputed missing review scores with zero to reflect any listings with zero reviews. We then standardized the price variable by removing currency symbols and converting values into a numeric format. Additionally, we reduced the impact of extreme values by removing listings in the top 1% of the price distribution, which could otherwise skew results. Overall, this process removed a small fraction of the data (about 0.8%) while maintaining proportional representation across the selected California markets, resulting in a clean and structured dataset.

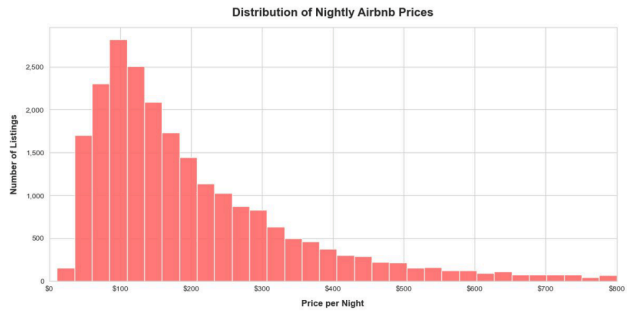
Next we conducted an exploratory analysis to better understand the structure of the Airbnb market across California. The first key insight on the distribution of listings clearly shows that the number of listings is highly uneven across the eight markets, with over 45,000 listings in Los Angeles. There are also several smaller markets such as San Mateo, Oakland, Santa Cruz, and Pacific Grove. This imbalance highlights that the Airbnb marketplace in California is heavily concentrated in a few major urban cities, which likely drives trends in pricing, demand, and availability. We then examined the types of accommodations available on the platform and found that the market is overwhelmingly dominated by entire home or apartment listings, with about 60,000 properties. This suggests that Airbnb is primarily used as a platform for renting private stays rather than shared accommodations, which could insinuate that guests are more likely to expect privacy and hosts should keep that in mind.

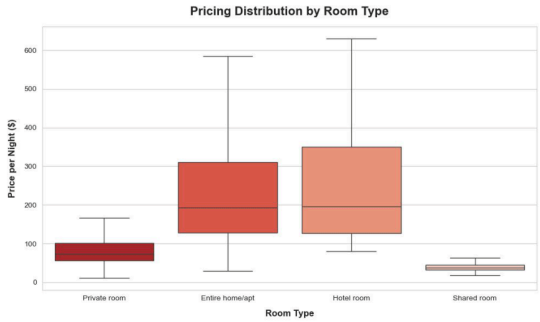
Next we analyzed the distribution of nightly prices across all listings and found a right-skewed distribution in the histogram, where the majority of listings are concentrated at lower price ranges ($50 - $200 per night). This skewness indicates that most of the Airbnb market operates in an affordable to mid-range pricing segment, while luxury listings still exist but are more rare. Finally, we explored how pricing varies by accommodation type using a boxplot. We found that entire homes and hotel rooms tend to have the highest median prices and a wider range of property sizes, amenities, and locations. Private rooms are significantly more affordable and show less variability, making them a more budget-friendly option for travelers. The overall

differences in pricing further support the idea that accommodation type is a major driver of pricing and should be a key feature in any predictive modeling.









<u>Q1: How do the key drivers of Airbnb listing demand vary across different locations in California?</u>

Model Performance Analysis - Linear Regression, Random Forest, and XGBoost

Objective

The objective of this analysis is to compare the predictive performance of three regression models: Linear Regression, Random Forest, and XGBoost on the Airbnb listings dataset. The target variable is reviews_per_month, which represents listing demand. The goal is to determine which modeling approach provides the most accurate predictions and to understand how model complexity affects performance.

Methodology

The cleaned dataset was divided into feature variables (X) and the target variable (y). Variables such as location, number_of_reviews, and number_of_reviews_ltm were excluded from the feature set to avoid leakage and redundancy. The data was then split into training and testing sets using an 80/20 train-test split. Three regression models were trained and evaluated: Linear Regression, Random Forest Regressor, and XGBoost Regressor.

Model performance was assessed using three standard evaluation metrics: R-squared (R^2), RMSE (Root Mean Squared Error), and MAE (Mean Absolute Error). R^2 measures the proportion of variance explained by the model, while RMSE and MAE capture the magnitude of prediction error.

Results

Model	R-squared	RMSE	MAE
Linear Regression	0.2666	1.4532	0.9111
Random Forest	0.5180	1.1781	0.5881
XGBoost	0.5302	1.1631	0.5942

The results show clear differences in predictive performance across the three models. Linear Regression produced the weakest results, with an R^2 of 0.2666. This means that the model explains only about 27% of the variation in reviews_per_month. Its relatively high RMSE and MAE also indicate larger prediction errors. This suggests that the relationship between the features and the target variable is not purely linear, and that a simple linear model is unable to capture the complexity of the dataset.

In contrast, both Random Forest and XGBoost performed substantially better than Linear Regression. The Random Forest model achieved an R^2 of 0.5180, reducing both RMSE and MAE considerably. This improvement shows that ensemble tree-based methods are better suited for this problem because they can model non-linear relationships and interactions among variables without requiring strong assumptions about the data.

Among the three models, XGBoost delivered the best overall performance, with the highest R^2 of 0.5302 and the lowest RMSE of 1.1631. Its MAE of 0.5942 was very close to that of Random Forest, showing that both models are strong performers. However, XGBoost provided the best overall balance across the evaluation metrics, making it the leading candidate at this stage of the analysis.

These findings establish a clear performance gap between the linear and ensemble-based approaches. While Linear Regression provides a useful benchmark, the stronger results from Random Forest and XGBoost suggest that model performance is closely tied to the ability to capture non-linear patterns and interactions within the data. In particular, the consistent advantage of XGBoost indicates that further refinement of model training and evaluation may yield additional improvements in predictive accuracy and reliability.

Cross-Validation and Hyperparameter Tuning Analysis

Objective

The objective of this analysis is to evaluate the predictive performance of multiple regression models using cross-validation and hyperparameter tuning. The purpose is to identify the model

that provides the best balance of explanatory power and prediction accuracy for the target variable.

Model Performance Comparison

The dataset was split into 64,738 training observations and 16,185 testing observations. A 10-fold cross-validation approach was applied to assess model stability and generalization, while final performance was evaluated on the test set using R^2, RMSE, and MAE.

The Linear Regression baseline achieved a mean cross-validated R^2 of 0.2702 and a test R^2 of 0.2666, indicating that it explains only about 27% of the variation in the target variable. Its RMSE of 1.4532 and MAE of 0.9111 were also substantially higher than those of the ensemble models. These results confirm that the relationship between the predictors and the target is not purely linear, and that a simple linear approach is unable to capture the complexity of the data.

The Random Forest baseline performed much better, with a mean cross-validated R^2 of 0.4968 and a test R^2 of 0.5180. It also reduced prediction error considerably, achieving an RMSE of 1.1781 and an MAE of 0.5881. This large improvement over Linear Regression shows that tree-based methods are far more effective for this problem because they can model non-linear relationships and interactions among variables.

The XGBoost baseline achieved a mean cross-validated R^2 of 0.5307 and a test R^2 of 0.5302, making it the strongest untuned model. Its RMSE of 1.1631 was lower than that of Random Forest, although its MAE of 0.5942 was slightly higher. Overall, XGBoost provided a stronger balance of fit and generalization, suggesting that boosting is especially effective for capturing subtle structure in the dataset.

Hyperparameter tuning produced mixed results. For Random Forest, the tuned model achieved a mean cross-validated R^2 of 0.4951 and a test R^2 of 0.5112, which was slightly lower than the baseline model. Its RMSE and MAE also increased marginally to 1.1863 and 0.5910. This suggests that tuning did not improve Random Forest performance in this case and that the baseline configuration was already close to optimal.

In contrast, XGBoost benefited from tuning. The tuned XGBoost model achieved the highest mean cross-validated R^2 of 0.5437 and the highest test R^2 of 0.5445 among all models. It also produced the lowest RMSE of 1.1452 and the lowest MAE of 0.5808, making it the best overall performer. These gains indicate that XGBoost is more responsive to parameter optimization and can better balance bias and variance when tuned carefully.

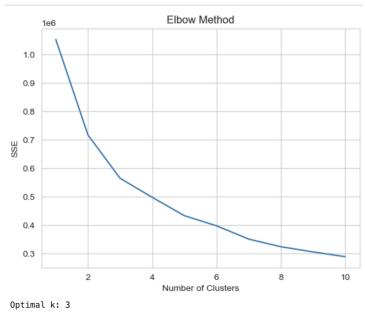
Overall, the results show a clear ranking in model performance. Linear Regression performed the weakest, Random Forest provided a major improvement, and XGBoost delivered the strongest results, particularly after tuning. This comparison demonstrates that more flexible ensemble methods are better suited for predicting the target variable, and that hyperparameter tuning is especially valuable when applied to XGBoost.

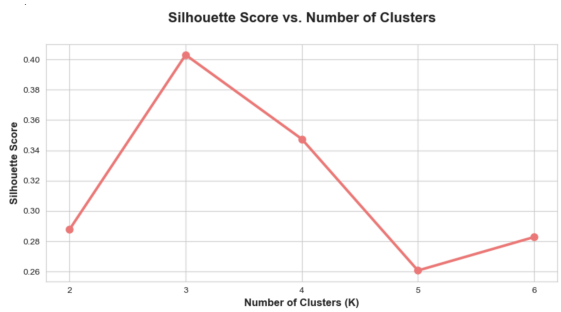
Q2: Can Airbnb listings be segmented into distinct market groups based on property characteristics, demand, and host behavior?

K-Means Clustering

To answer the second question, we have used two different clustering methods: K-Means, and DBSCAN. We understand that K-Means is better suited for the Airbnb dataset because we are trying to create interpretable clusters and K-Means will partition all data into K groups and also produce centroids that show the average profile. We are only incorporating DBSCAN to compare and contrast the results with K-Means, and potentially identify any outliers or noise.

For K-Means, we first selected features that represent property characteristics, demand, and host behavior, this includes variables like number of bedrooms, bathrooms, people accommodated, price, availability, number of reviews, ratings, and number of listings the host has. Then, we imputed missing values using K-Nearest Neighbors (KNN) so that missing values are filled with listings that are similar. Next, we standardized the features using a standard scaler, which transforms each variable and ensures that all features are on the same scale before applying K-Means clustering. We used both the Elbow Method and the Silhouette Method to identify the optimal K value. Both methods indicated an optimal k = 3, where the elbow point occurs and the silhouette score is highest. Therefore, we used 3 clusters, and each observation was assigned to a cluster. Finally, the characteristics of each cluster were analyzed by computing the mean values of all numerical features within each cluster.





From the cluster characteristics above, we identified the features that most strongly characterize and distinguish each cluster to create a cluster profile as shown below.

Cluster	Segment Name	# of Listings in Cluster	% of Total Market	Price	Accommodates	Bathrooms	Bedrooms	Availability (30 days)	Rating Score	Host # Listings in Area
0	Typical / Mainstream	48973	60.5%	150.84	2.99	1.19	1.25	12.74	4.80	13.30
1	Luxurious / Premium	13561	16.8%	515.94	8.64	2.90	3.71	15.12	4.52	15.63
2	Commercial Hosts	18389	22.7%	227.81	3.54	1.48	1.63	16.51	0.03	46.61

We can see that the market is dominated by Cluster 0, accounting for 60.5% of the market, this represents typical mainstream hosts which are the most common listings people would find. The average price for this cluster is 150.84 which is the lowest of all, and could perhaps reveal why

people choose Airbnb for a standard room rather than commercial hosts and hotels. The typical listing accommodates 2.99 people, and has a rating of 4.80 which is very high.

Cluster 1 consists of luxurious and premium listings, making up the least in the market with 16.8%. These listings have a much higher average price of 515.94. The listings accommodate the most people (8.64 people), have the most bedrooms and bathrooms, the rating score of 4.52, and each host has an average of 15.63 listings which is slightly higher than a typical host.

Cluster 2 consists of commercial hosts, making up the second highest of the market with 22.7%, which could include property management companies, apartments, and hotels. This is because the hosts have an average of 46.61 listings in the area and 438.56 in total, which would be very high for an individual host. These listings have an average price per night of 227.81, accommodate 3.54 people, and have an average rating score of 0.03. However, this is worth noting that the near 0 rating does not mean a very low rating, it could simply be due to imputing missing reviews, thus values being filled with low or near-zero values and lowers the entire average.

The results reveal that a typical listing and luxurious listings have very similar features such as number of beds/bathrooms, but luxurious listings have a higher price point, which could be due to other factors not included in this analysis. Commercial hosts make up a significant proportion of the Airbnb market, but do not receive a lot of reviews which contributes to a limitation of this analysis.

Clustering: DBSCAN

To explore whether airbnb listings can be segmented into distinct market analysis, we also applied DBSCAN. Unlike K-means, DBSCAN groups observations based on data density rather than the distance to cluster centroids which allows DBSCAN to detect noise points (outliers) such as unusual listings incorporated into typical market segments.

The same set of features used in the K-Means analysis was selected, and missing values were handled using KNN imputation. All features were standardized using a StandardScaler to ensure comparability across variables. DBSCAN was implemented with parameters eps = 2 and a minimum sample size of 10. The model was fitted on the scaled dataset, and cluster labels were assigned to each observation.

Points labeled as -1 were treated as noise. To evaluate clustering quality, the number of clusters, proportion of noise points, and silhouette score (excluding noise) were computed. To visualize the clustering structure, we projected both K-Means and DBSCAN into PCA for easier comparison.

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DBSCAN clusters: 2
Noise points: 436 (0.5%)
-1      436
 0    80463
 1       24
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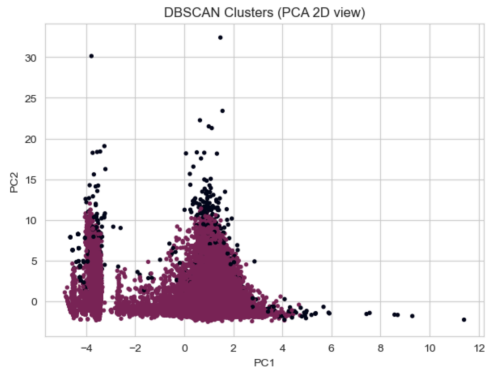
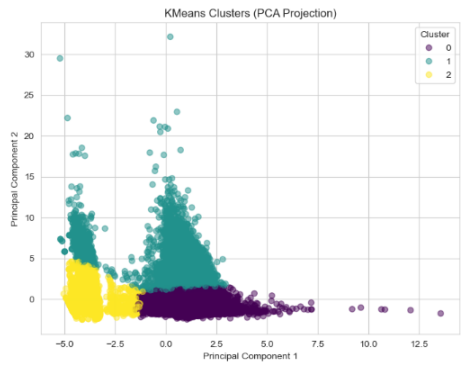




The results show that DBSCAN identified only 2 clusters, with approximately 0.5% of observations classified as noise. The vast majority of listings were grouped into a single dominant cluster (80,463 observations), while the second cluster contained only a small subset of listings. Although the silhouette score (approximately 0.597) suggests moderate separation, the clustering structure is highly imbalanced and lacks meaningful segmentation. The PCA visualization further confirms that DBSCAN fails to clearly separate distinct groups, with most points concentrated in overlapping regions.

As we expected, DBSCAN is not suited due to the continuous and high-dimensional nature of the Airbnb data. The lack of clear density boundaries results in poor cluster differentiation, making DBSCAN ineffective for identifying actionable market segments in this context.

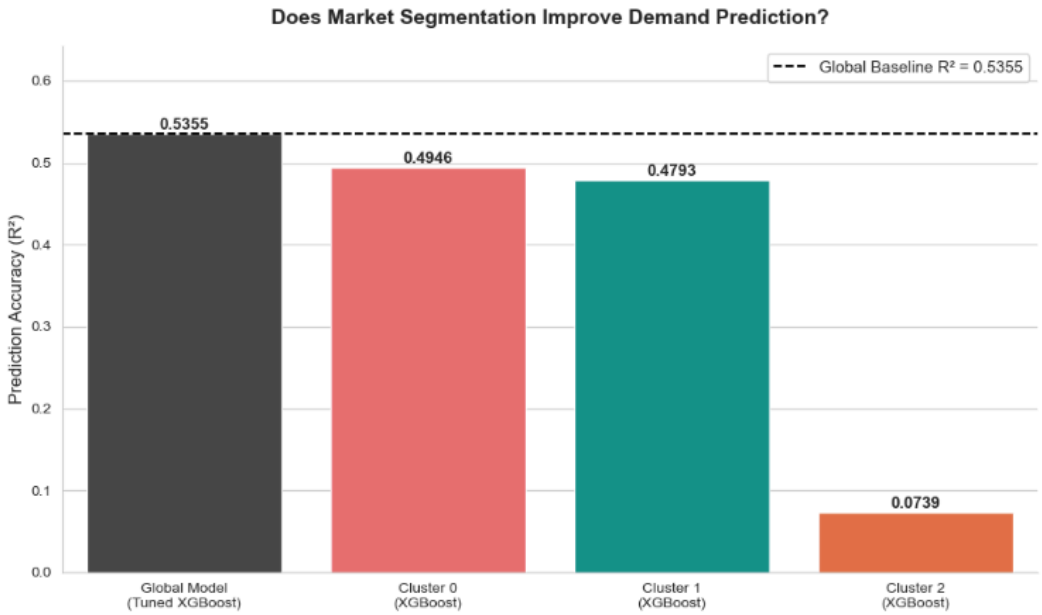
<u>Predictive Question Analysis: Can clustering-based market segmentation improve the prediction of Airbnb listing demand across California?</u>

To answer our main predictive question, we evaluated whether clustering-based segmentation improves the prediction of Airbnb listing demand.

First, the dataset was filtered to include only active listings, ensuring that the analysis focuses on relevant observations with positive demand. We selected the same set of features used in clustering including property characteristics, demand indicators, and host behavior. Missing values were handled using the previously fitted KNN imputer to maintain consistency. The data was then standardized using the same scaler before assigning each listing to a cluster using the trained K-Means model.

Next, a global baseline model was built using XGBoost to predict listing demand across all active listings. XGBoost was chosen as it has the best overall performance , achieving the highest R^2 and lowest RMSE and MAE. This model excluded non-predictive variables such as location identifiers, cluster labels, and redundant review metrics. The dataset was split into training and testing sets, and model performance was evaluated using R^2. This baseline serves as a benchmark to compare against cluster-specific models.

Finally, separate XGBoost models were trained for each cluster to assess whether segmentation improves predictive accuracy. For each cluster, the data was filtered, split into training and testing sets, and evaluated using the same methodology as the global model. The results were then compared to determine whether specialized models outperform the global approach, providing insight into the effectiveness of clustering-based segmentation for demand prediction.



Conclusion

This analysis examined whether clustering-based market segmentation can improve the prediction of Airbnb listing demand across California. While clustering provides meaningful insights into market structure, the results highlight important trade-offs between segmentation and predictive performance. Our analysis shows that a global "one-size-fits-all" model is more effective at capturing overall demand patterns, whereas segmentation reveals underlying differences across distinct groups of listings.

Using K-Means clustering, we identified three segments representing different types of Airbnb listings. However, when applying XGBoost models within each cluster, we observed that cluster-specific models did not outperform the global baseline model. The global model achieved an R^2 of 0.5355, while Cluster 0 ($R^2 = 0.4946$) and Cluster 1 ($R^2 = 0.4793$) both underperformed relative to the baseline. Cluster 2 performed significantly worse ($R^2 = 0.0739$), indicating very low predictive power within this segment. These results suggest that while segmentation allows for more targeted analysis, it may reduce model performance due to smaller sample sizes or weaker predictive signals within individual clusters.

A key insight from this analysis is the variability in predictability across clusters. Clusters 0 and 1 demonstrate moderate predictive performance, whereas Cluster 2 is highly unpredictable. This likely reflects the presence of commercial hosts or irregular listing behaviors that are not well captured by the available features. As a result, certain segments of the market may depend on external factors not included in this dataset.

Overall, the findings indicate that Airbnb is not a homogeneous market. While clustering enhances understanding of market dynamics and enables more tailored strategic insights, it does not necessarily improve predictive accuracy. For demand forecasting, global models remain more effective, whereas segmentation is better suited for interpreting market structure and informing targeted decision-making.